THE SI SENTINEL

Weekday briefing on builders shaping super intelligence

THURSDAY, OCTOBER 29, 2026

THE MORNING PICTURE

Google ships Gemini 4 Carbon two weeks after Argon, and Alphabet and Microsoft post AI-driven quarters

Google introduced Gemini 4 Carbon on Wednesday, a frontier model that tops the new version of the Artificial Analysis Intelligence Index and leads on cybersecurity tests. Hours later Alphabet reported a half-trillion-dollar annual revenue run rate and 103% growth at Google Cloud. Microsoft followed with $92.3 billion in quarterly revenue, Azure up 48% and a $677 billion backlog of commercial contracts.

Elsewhere, GitHub launched cheaper Copilot Fusion models, Vercel opened an agent-ready build service, and Suhail Doshi unveiled Particle, a developer-focused model company. Shopify put Shop Pay inside AI agents, including Grok Bot, and Cartesia showed a byte-level model that skips the tokenizer. SpaceX finished static fires on the Starship that Elon Musk says will make the first ship tower catch.

I · FRONTIER MODELS

Gemini 4 Carbon takes the top spot two weeks after Argon

Google's newest model leads Artificial Analysis's rebuilt index and three security benchmarks. Trusted cyber defenders get it first, with a gradual rollout to follow.

59	88.2%	$2 / $10
Score on the Artificial Analysis Intelligence Index v5.0, ranked first	On SEC-bench Pro, which tests finding software vulnerabilities	Introductory price per million input and output tokens

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

"I did say we'd be releasing our models periodically," Pichai wrote at 1:01pm EDT Wednesday, introducing Gemini 4 Carbon as "our next frontier model." He said it ranks first on version 5.0 of the Artificial Analysis Intelligence Index at 59, and scored 88.2% on SEC-bench Pro for finding vulnerabilities, 54.8% on ExploitGym for exploitation and 71.0% on CWE-bench for patching. "We're seeing significant improvements in long-horizon tasks even compared to Argon," he added, and "some trusted cyber defenders will get access starting this week," with more partners to follow through Google's Fairwind Program "ahead of wider availability soon."

Google's announcement, by Koray Kavukcuoglu, said Carbon went through the US government's voluntary pre-release testing. It lists an introductory price of $2 per million input tokens and $10 per million output tokens, rising later to $4 and $20, with cached input 95% cheaper. The model placed fourth on WebDev Arena at 1,768, up 89 points. Google said Carbon ran "for days" on engineering tasks, and that the Chrome team saw patches more than 10 times better than those from another lab's model.



Google's main campus in Mountain View, California, seen from the air in 2013.
PHOTO: AUSTIN MCKINLEY · CC BY 3.0 VIA WIKIMEDIA COMMONS

Safety numbers Google's evaluation results

Google reported a 0.0% cheating rate on Terminal-Bench 4.0, high-severity misalignment alerts down about 80% and

unsupported claims roughly halved. On Gray Swan's prompt-injection test Carbon scored 0.0%, against 0.7% for Argon, and it more than doubled Argon's secure-code gains, according to the post.

Demis Hassabis @demishassabis · CEO, Google DeepMind

Hassabis, whose lab built the model, posted two minutes after Pichai:

> *"Iterating at lightning speed! Gemini 4 Carbon is SOTA across key benchmarks and combines frontier capabilities with deep safety & security research."*
>
> DEMIS HASSABIS · @DEMISHASSABIS · 1:03PM EDT · POST

SOTA means state of the art. Elon Musk replied "Impressive" to Hassabis at 1:50pm EDT and quoted Pichai's post with the same word at 1:59pm EDT. Hassabis answered "Thanks!"

II · EARNINGS

Alphabet and Microsoft say AI demand is lifting both clouds

Google Cloud more than doubled its revenue. Microsoft's commercial backlog rose 73%, and its quarterly spending on property and equipment roughly doubled.

103%	$92.3B	48%
Google Cloud revenue growth in Alphabet's third quarter	Microsoft revenue for its fiscal first quarter, up 19%	Azure growth, with Microsoft Cloud at $63.9 billion

Sundar Pichai @sundarpichai · CEO, Google and Alphabet

Pichai summed up the quarter at 4:05pm EDT:

> *"Q3 is in the books! We're now on a half-trillion annual revenue run rate, with Alphabet revenues growing at 25%. - Our AI investments are driving overall Search query growth, including commercial queries - our YoY query growth rate is the highest it's been in the last five years. - Google Cloud revenue grew 103%, powered by extraordinary demand for our AI infrastructure and AI solutions. Our 1P model APIs now process nearly 32B tokens/min (up 40% QoQ). - Strong momentum in model progress with Argon and Carbon leading the frontier, and Genie 4 our most advanced new world model yet."*
>
> SUNDAR PICHAI · @SUNDARPICHAI · 4:05PM EDT · POST

His full remarks said the Gemini app has reached 1 billion users and that Argon's first-week usage grew 3.5 times faster than Gemini 3's. Google teams now ship 35% more code with 30% fewer errors, he said. Waymo has passed 300 million autonomous miles in 15 US cities, has started service in San Antonio, San Diego and Tampa, and plans to launch in Canada, Germany, Singapore and Spain. Pichai said he expects progress to speed up over the next 12 months "because of RSI," or recursive self-improvement. Musk's reply at 10:23pm EDT: "Wow."

Satya Nadella @satyanadella · CEO, Microsoft

Microsoft's results for its fiscal first quarter showed revenue of $92.3 billion, up 19%, operating income of $45.9 billion, up 21%, and net income of $36.6 billion, up 32%, or $4.92 a share. Microsoft Cloud revenue reached $63.9 billion, up 30%, and commercial remaining performance obligation, the backlog of contracted revenue, rose 73% to $677 billion. Additions to property and equipment were $39.4 billion, against $19.4 billion a year earlier.



Building 92, the visitor center on Microsoft's campus in Redmond, Washington, in June 2015.

PHOTO: JIAQIAN AIRPLANEFAN · CC BY 3.0 VIA WIKIMEDIA COMMONS

"It was a fast start to our fiscal year," Nadella wrote at 6:53pm EDT, opening a thread on the call. He said Copilot daily users "grew nearly 50% QoQ, and more than 5X YoY," and customers with more than 50,000 Copilot seats rose eightfold. Foundry revenue "more than tripled YoY" and revenue from open-source models rose tenfold. More than 50 million agents now run in Agent 365, and he said OpenAI Codex, Anthropic Claude Code and Meta Muse are adopting MXC, Microsoft's secure on-device boundary for agents. "I have never been more confident in the opportunity ahead of us," he closed. Musk replied "Impressive."

ON THE WIRE

Waymo's miles. Alphabet's robotaxi unit has passed 300 million autonomous miles in 15 US cities, Pichai said in his remarks.

GitHub, Vercel and a new lab compete on cheaper, faster agent work

Copilot Fusion promises frontier quality at a fraction of the cost. Vercel wants build pipelines that agents can run, and Particle trims the tokens a coding model spends.

77%	2–6×	~45%
Lower cost Copilot Fusion can offer, per GitHub	Faster builds Vercel saw in its own repositories	Fewer tokens particle-1 uses at the same performance

Satya Nadella @satyanadella · CEO, Microsoft

At GitHub Universe, Nadella wrote at 1:08pm EDT that Microsoft is redefining the software development lifecycle "for the agentic era." GitHub Copilot Fusion offers "frontier-level quality, 40% faster and at up to 77% lower cost," he said. The Copilot harness is now in Foundry, and Copilot Projects coordinates agents across repositories. GitHub's post by Jay Parikh said Fusion Max scored within two points of Opus 5.5 Max on the Artificial Analysis Coding Agent Index, at half the cost and in 40% less time. Enterprise customers now get every Copilot feature without a separate subscription, paying for AI credits. The post also predicts that "all the software we've written to date will amount to less than 1% of the software we'll write in the next five years."

ON THE WIRE

Leaner harness. Vercel's fx version 0.0.15 uses 70% less memory in long sessions.

Guillermo Rauch @rauchg · CEO, Vercel

Vercel opened Vercel CI, its continuous-integration service, in beta on all plans. It uses Turborepo to run only the tasks a change affects, and it was two to six times faster in Vercel's own repositories. Rauch said at 4:21pm EDT that the company runs "20 worldwide clusters doing millions of microVMs per day," and that a command-line tool lets agents "inspect, abort, trigger, replay CI." His conclusion: "Agentic CI is the future of how we ship software." Later Vercel added virtual models to its AI Gateway: one alias that a team can point at whichever model it wants.

> *"This is like DNS for models. An alias that *resolves* to a model, giving you traffic-steering abilities. Solves a huge number of problems. From not having teams stuck on old model versions, to saving on costs, to removing "decision paralysis". Try it!"*
>
> GUILLERMO RAUCH · @RAUCHG · 6:38PM EDT · POST

Vercel's open-source agent harness, fx, also shipped version 0.0.15, which uses 70% less memory in long sessions, and became a built-in provider in Theo Browne's T3 Code. Rauch's verdict at 2:02pm EDT: "Theo cooked."

QUICK HIT

Uncle Bob Martin (@unclebobmartin), author of "Clean Code," at 10:43am EDT Wednesday, answering a developer who asked why people should read programming books if agents write the code: "As a manager you have to check the work of the entities you manage. That means you need to know the principles by which they should operate. And that means you'll have to do regular spot checks. How often you do them is a function of how much trust you've developed over the history of your interactions. But trust is never absolute. Trust. But Verify."

Suhail Doshi @Suhail · Founder, Particle

Doshi, who co-founded Mixpanel, announced Particle at 1:36pm EDT as "our new company," starting with a series of models for developers. He said particle-1 keeps "parity in performance while using ~45% fewer tokens." Particle's launch post said the model is post-trained on DeepSeek V4.1 Flash, has a context window of 1 million tokens, and uses 28.4% to 52.2% fewer output tokens per task, at 34.4% to 51.6% lower cost. Its backers include Y Combinator, 8VC, Patrick Collison and Adam D'Angelo. The model is available on Vercel's AI Gateway, and Rauch called the partnership a way to make intelligence "as accessible and abundant as electricity."

Aaron Levie @levie · CEO, Box

Levie saw the launch as part of a bigger shift in demand:

> *"Agents are going to consume 1,000X more tokens than we consume today. Because they're going to be running in our workflows, processing all of our data, building out nearly all software, securing enterprises, and much more. To do this at scale we will need a variety of approaches at the frontier of AI. This includes continuing to push the boundaries of closed models, while also increasing the capability of open weights models, making them more secure, as well as making them more efficient for large workloads. Very exciting to see what @Suhail is building with Particle on this front."*
>
> AARON LEVIE · @LEVIE · 2:49PM EDT · POST

David Heinemeier Hansson @dhh · Creator, Ruby on Rails

"Basecamp for Claude is live!" Heinemeier Hansson wrote at 6:43pm EDT. The connector lets Anthropic's assistant work

inside the project tool his company makes. He also cheered a plan to run his Omarchy Linux setup on an iPad: "Yes. Yes. YES!"

IV · AGENT COMMERCE

Shop Pay moves into AI agents, and Musk tells shoppers where to buy

Shopify's fall Edition lets agents pay at stores that are not on Shopify. Grok Bot added Shop Pay the same night, and Meta's Muse kept climbing the app charts.

400M	1B+	4.2×
People who use Shop Pay, per Tobi Lütke	Products in the Shopify Catalog that agents can search	Year-over-year growth in daily Sidekick sessions

Tobi Lütke @tobi · CEO, Shopify

"The new Edition drops today. It's a magical time to build," Lütke wrote at 12:10pm EDT, introducing Shopify's Fall '26 Edition. Shop Pay now works inside personal agents and at stores off Shopify: "400 million people use Shop Pay and now it's in the hands of developers everywhere." Agent builders get Shop sign-in, the Shopify Catalog and Shop Pay at shop.dev, and Lütke said the company is "starting to add products from brands that aren't on Shopify" to the Catalog. Shopify's page lists Shop Pay inside Muse, Claude Duet, Grok Bot and Instinct.

The Sidekick assistant can now "stage multistep changes", and stores built with Canvas launch 20% faster. The Shop app checks out from several brands at once. Campaign Autopilot is open to all merchants, and an Expo Mode was built with Google's new HTML-in-Canvas API.

Noah Shinn @noahrshinn · Instinct

Shinn said at 3:11pm EDT that Instinct's agent "can use Shop Pay at any online store," adding: "You approve every purchase in Shop Pay, and your card details never leave Shop Pay." Lütke replied "So good."

Elon Musk @elonmusk · CEO, Tesla and SpaceX

Grok Bot version 0.78.1 rolled out at 7:27pm EDT with shopping results shown as product cards and a way to connect Shop Pay in its vault, according to the changelog. Lütke quoted the release at 10:30pm EDT: "Let's go!" Musk quoted a user who said Amazon locked out his Grok Bot, so it "just bought the same thing from a different site (at a lower price)":

> *"That's what I do too. Just have @Bot buy it somewhere else at a lower price."*
> ELON MUSK · @ELONMUSK · 10:30PM EDT · POST

Lütke answered "Just in time for shop pay support," and Musk quoted him eight minutes later: "Buy from @Shopify!"

Alexandr Wang @alexandr_wang · Chief AI Officer, Meta

Wang said at 8:27pm EDT that the "muse global tour" was "going pretty well." He was quoting Appfigures data showing Muse above ChatGPT on the iPhone free chart in all four countries where it is live: first in Singapore, Japan and Canada, and third in the US against ChatGPT's fourth. An hour later he pointed to Artificial Analysis's personality benchmark, which rated Muse models the most "Friendly" of all models.

V · RESEARCH

Byte-level models, drug discovery and an open Mistral

Cartesia says a model without a tokenizer can match a conventional one with far less data. Anthropic described early drug results, and Mistral threw launch parties for an open model.

Albert Gu @_albertgu · Co-founder, Cartesia

Gu, a Carnegie Mellon professor, said at 1:53pm EDT that his team's new H-Net, which reads raw bytes instead of tokens, "finds up to 3-5x efficiency gains in controlled scaling law experiments, and at larger scale matches an open-source model trained on 11T tokens with ~10x less data and compute." Cartesia's write-up said a 3-billion-parameter H-Net matched SmolLM3's average across six tasks, and that the architecture will be released. Will DePue called it "real and a big deal," noting that "the ugliest part of the transformer is the tokenizer." Musk replied to DePue at 10:15pm EDT: "Bitstream in, bitstream out."

Anthropic @AnthropicAI · Drug development

Anthropic reported early results from six drug-development partners, including Bristol Myers Squibb and Generate Biomedicines. Divergence Labs produced about 35,000

antibody designs in a month. BigHat Biosciences tested about 1,010 variants in 10 weeks, with more than 10 times tighter binding. At Enveda, Claude caught all 11 issues found by human reviewers plus seven more, cut review time by 89% and drafted a regulatory document in four hours instead of about three months. Anthropic also acquired Mithril Medicine, which is working on muscular dystrophy, and said a human trial could begin as soon as late 2028. Musk replied "Interesting" at 11:04pm EDT.

QUICK HIT

Elon Musk (@elonmusk), at 11:03pm EDT Wednesday, quoting a post that said "a massive shift" in human biology is "sitting in press releases and almost nobody reads them": "If you look closely enough, breakthroughs in everything are accelerating exponentially. An exponential always starts off looking small."

Guillaume Lample @GuillaumeLample · Co-founder, Mistral AI

Mistral is celebrating its newest large model with parties in three cities. Lample's announcement:

> *"This Thursday, we are hosting events at our SF, NYC, and London offices to celebrate the launch of ML4 ! We're growing our teams across all three locations, as well as in Paris, Warsaw, Zurich, and Montreal, where we recently opened a new office. If you believe the future of AI should be open, that scientific breakthroughs should be shared, and that everyone should benefit from frontier intelligence, come build that future with us !"*
>
> GUILLAUME LAMPLE · @GUILLAUMELAMPLE · 8:12PM EDT · POST

Mistral Large 4 entered public preview on Oct. 6. It has about 1 trillion parameters, 52 billion of them active at a time, and was trained on Grace Blackwell chips in Mistral's European data centers. Mistral promised open weights "by the end of the month," and its Hugging Face page still lists them as an upcoming release.

Vitalik Buterin @VitalikButerin · Co-founder, Ethereum

Replying at 12:11am EDT Thursday to writer Ruxandra Teslo, who called AI commentary on her essays "still at the level of slop," Buterin wrote that he finds AI "extremely knowledgeable, and so exceptionally able to detect references to almost anything, but not very wise." Its knowledge is easier to see than its limited wisdom, he added: "it's perhaps even the very definition of wisdom that it cannot be easily verified."

John Carmack @ID_AA_Carmack · Founder, Keen Technologies

Carmack kept up his bet that robots will not handle everyday chores within a couple of years. He offered a test at 6:37pm EDT: "Go change the batteries in the jack-o-lantern out front." "An open source LLM can trivially understand this," he wrote, but he doubts an open vision-language-action model and "a sufficiently dexterous robot" could do it "in, say, an hour of trying. Still want to bet?"

VI · SPACE AND AUTONOMY

Ship 42 finishes its static fires, and Musk names it for the first catch

The Starship for Flight 15 fired all six engines and relit one to rehearse a deorbit burn. On the ground, Tesla's robotaxi area grew across Houston and Starlink kept Spanish phones online.

Elon Musk @elonmusk · CEO, Tesla and SpaceX

SpaceX said at 12:10pm EDT Wednesday that Ship 42's static fires are complete: the Starship for Flight 15 "performed a full duration, six-engine static fire, and fired a single Raptor engine to simulate a deorbit burn." SpaceX has not announced a launch date. Musk quoted the update that night:

> *"As the prophecy foretold, Starship 42 will be the first tower catch"*
>
> ELON MUSK · @ELONMUSK · 10:37PM EDT · POST

SpaceX has caught Super Heavy boosters with the launch tower's arms but has never caught a ship. Replying to a clip on why Starship is built from stainless steel, Musk added at 2:20am EDT Thursday: "Also, no need to paint stainless. Paint adds more mass & cost."



A Super Heavy booster descends to the tower's arms at Starbase, Texas, for the first booster catch during Starship's fifth flight test on Oct. 13, 2024.
PHOTO: STEVE JURVETSON · CC BY 2.0 VIA WIKIMEDIA COMMONS

Robotaxi Houston Tesla

Houston police said Tesla is "expanding their Cybercab/Robotaxi services area across #Houston," giving

people "more ways than ever to get home safely after a night out." Musk quoted them at 1:49pm EDT: "Robotaxi Houston." Earlier he shared the story of a 98-year-old owner whose self-driving Model Y stopped for a chipmunk crossing the road: "At Tesla, we even care about the chipmunks."

Starlink @Starlink · SpaceX

Starlink said its mobile service is providing free connectivity to Orange and Telefónica customers affected by the train collision in Spain, so compatible phones can use apps and texts "even when cell towers are down." Musk's comment at 11:06pm EDT: "Standard practice."

VII · PLATFORMS AND FACTORIES

Sweeney presses Apple, and Luckey says American manufacturing can come back

Epic's chief executive called App Store review a dictatorship and defended Steam's openness to AI-made games. Anduril's founder called cost estimates for US-built electronics a cover story for outsourcing.

Tim Sweeney @TimSweeneyEpic · CEO, Epic Games

Responding to a post that mocked a rejected developer at 6:32am EDT Thursday, Sweeney asked why Apple should be able to "arbitrarily deny this developer access to a billion users." "I started out making programs on an Apple][and sharing them with friends. Apple-as-app-dictator is disgraceful," he wrote. A minute later he added that if Apple wants to stock only apps it chooses, "fine, but stop blocking or crippling all competing stores worldwide." Epic has won fights to open app stores "in EU, Japan, and Brazil," he noted on Wednesday night.

On games made with AI, he argued that critics "aren't going to take on Steam" because its users would "out-mob their mob."

QUICK HIT

Elon Musk (@elonmusk), at 2:04am EDT Thursday, replying to a post about justifying spending with "a future of abundance": "Saving money for the distant future is pointless with the advent of super intelligent abundance"

Palmer Luckey @PalmerLuckey · Founder, Anduril

Luckey quoted a clip of himself saying that analysts who claim an American iPhone would cost $37,000 are "trying to prop up a status quo that the megacorporations want to keep going." His caption at 12:12am EDT Thursday: "Never, ever let them tell you it can't be done." Two minutes later, replying to a post calling domestic production "a national security imperative," he wrote: "Yes! America can do it, and so we must!"

Takeaways

1. Google now ships frontier models about every two weeks, and Carbon reaches cyber defenders first.
2. AI demand is showing up in cloud revenue: 103% growth at Google Cloud and 48% at Azure.
3. The coding race has turned to cost per task, with cheaper Copilot models and Particle's leaner tokens.
4. Payment rails are being rebuilt for agents, and Shop Pay now works at stores off Shopify.
5. Byte-level models could make the tokenizer optional if Cartesia's results hold at scale.
6. Ship 42 is lined up for the first attempt to catch a Starship upper stage.

On the beat

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Elon Musk, Tobi Lütke, Tim Sweeney, Uncle Bob Martin, Palmer Luckey, Guillermo Rauch, Satya Nadella, Aidan Gomez, David Heinemeier Hansson, Paul Graham, Alexandr Wang, Sundar Pichai, Demis Hassabis, Michael Dell, Vitalik Buterin, Guillaume Lample, Kevin Weil, Joe Gebbia, John Carmack, Aaron Levie.

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Marc Andreessen, Jensen Huang, Ashok Elluswamy, Andrej Karpathy, Sam Altman, Dario Amodei, Patrick Collison, Yann LeCun, Jeff Dean, François Chollet, Fei-Fei Li, Greg Brockman, Mustafa Suleyman, Andrew Ng, Michael Truell, Tom Brown, Clément Delangue, Jim Fan, Mira Murati, Mark Zuckerberg, Noam Brown, David Singleton, Ilya Sutskever, Lisa Su, Kanjun Qiu, Nat Friedman, Arthur Mensch, Mark Chen.

Added Suhail Doshi (@Suhail), founder of Particle and co-founder of Mixpanel, and Albert Gu (@_albertgu), co-founder of Cartesia and a Carnegie Mellon professor. The list now stands at 50.

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